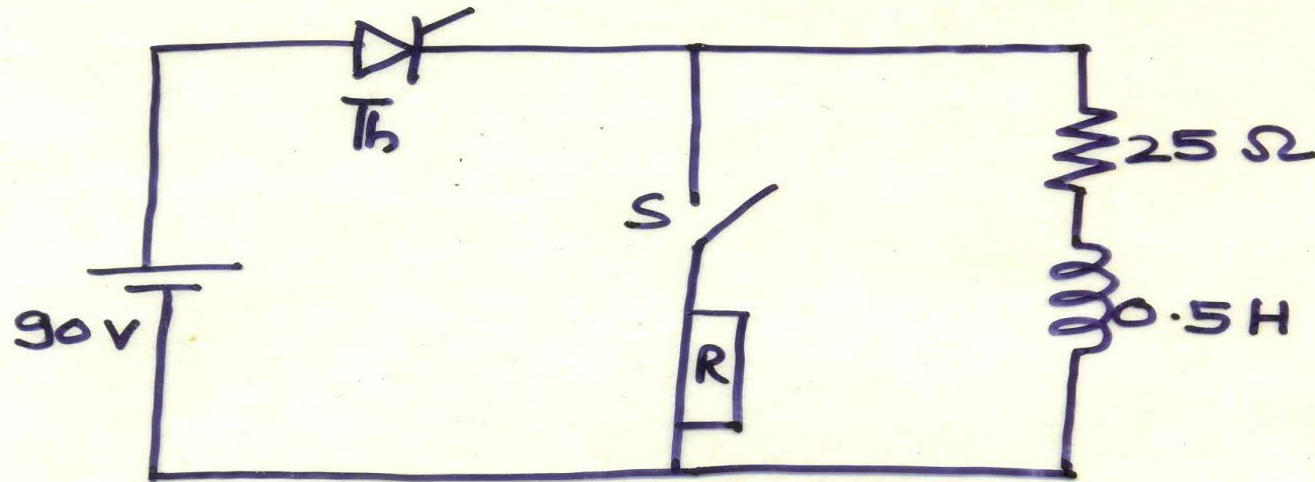


PE Numericals Unit II

Tut 3 Q 3

The thyristor shown in figure has a latching current of 40 mA & is fired by a pulse of width $40\text{ }\mu\text{s}$. Find if the thyristor will turn on. If the switch is closed, find the maximum value of R so that thyristor may turn on.



Tut 6 Q 3

Q.3.) A single phase half wave rectifier feeds a 15Ω resistance. The supply voltage is 230 V (rms) . The firing angle is $\pi/2$.

Find

- (a) V_{dc} , I_{dc} , V_{rms} , I_{rms} .
- (b) Rectification efficiency.
- (c) Form Factor & Ripple Factor.
- (d) VA rating of transformer feeding the rectifier.
- (e) TUF
- (f) PIV across thyristor.

Tut 7 Q 2

Q.2.) A single phase full converter bridge circuit is feeding an RLE load & is fed from 230 V, 50 Hz single phase mains. The load current is constant at 15 A. $R = 0.5 \Omega$, $L = 0.3 \text{ H}$.

Find :-

- (a) firing angle if $E = 100 \text{ V}$.
- (b) firing angle if $E = -100 \text{ V}$.
- (c) input pf in both cases.

Tut 7 Q 3

Q.3.) A single phase fully controlled bridge converter feeds an RL load. The input voltage is 230 V & firing angle is 30° . The load current may be assumed to be constant at 10 A.

Find:-

- (a) DC output voltage.
- (b) RMS value of input current.
- (c) Fundamental component of input current.
- (d) Displacement Factor.
- (e) Input Power Factor.
- (f) Harmonic Factor.
- (g) Ripple Factor.

Tut 8 Q 1

A three phase full converter supplies a battery from three phase 230 V 50 Hz mains. Battery emf = 200 V, battery internal resistance = 0.5Ω . Current has a constant value of 20 A. Find firing angle and input power factor.